

Write in Terminal

To make a new file:

```
touch file.txt
```

Write there:

```
nano file.txt
```

Save that:

```
Ctrl +o then press enter
```

Exit:

```
Ctrl + X
```

Search and Filtering

Find:

```
find . -name "*.txt"
```

Search Inside files:

```
grep "error" file.txt
```

Replace all contents and add new:

```
echo "new content" > filename.txt
```

◆ 1 File Searching & Filtering

1) **find** — Advanced File Search

Description: Search files based on name, type, size, permission.

```
find . -name "*.txt"
```

Search by size:

```
find . -size +1M
```

Search by permission:

```
find . -perm 644
```

Search and delete:

```
find . -name "*.tmp" -delete
```

2) **grep** — Search Inside Files

Search word inside file:

```
grep "error" file.txt
```

Case insensitive:

```
grep -i "linux" file.txt
```

Search recursively:

```
grep -r "main" .
```

◆ **2** Redirection & Pipes (Very Important for OS)

3) Output Redirection

Overwrite:

```
echo "Hello" > file.txt
```

Append:

```
echo "World" >> file.txt
```

Error redirection:

```
ls wrongfile 2> error.txt
```

Redirect both:

```
ls wrongfile > all.txt 2>&1
```

4) Pipes |

Send output of one command to another:

```
ls -l | grep ".txt"
```

Count lines:

```
ls | wc -l
```

◆ **3** Process Management (Core OS Concept)

5) **ps** — Show Running Processes

```
ps  
ps aux
```

6) **top** — Real-Time Process Monitor

```
top
```

Press **q** to exit.

7) **kill** — Terminate Process

```
kill PID  
kill -9 PID
```

Find PID first:

```
ps aux | grep processname
```

◆ **4** Disk & Storage Commands

8) **df** — Disk Free Space

df -h

9) **du** — Disk Usage

du -sh *

◆ **5** Permissions & Ownership Advanced

10) **umask**

Shows default permission:

umask

Set new mask:

umask 022

11) **chattr** (Advanced Attribute Control)

Make file immutable:

sudo chattr +i file.txt

Remove:

sudo chattr -i file.txt

◆ **6** Archiving & Compression

12) **tar**

Create archive:

tar -cvf archive.tar dir1

Extract:

```
tar -xvf archive.tar
```

With compression:

```
tar -czvf archive.tar.gz dir1
```

◆ **7** Networking Basics

13) **ping**

```
ping google.com
```

14) **ifconfig** (or **ip** modern command)

```
ip a
```

◆ **8** System Information

15) **uname**

```
uname -a
```

16) **whoami**

```
whoami
```

🔥 **1** Inode & File System Internals

1) **ls -i** — Show Inode Number

```
ls -i file.txt
```

Every file has an **inode number** (metadata structure in Linux).

Check inode for all files:

ls -li

2) **stat** — Detailed File Metadata

stat file.txt

Shows:

- Inode number
- Size
- Blocks
- Access time
- Modify time
- Permissions

Very useful for understanding file system structure.

2 Hard Link vs Soft Link (Important OS Topic)

3) Hard Link

In file.txt hardlink.txt

Check inode:

ls -li

You'll see both share same inode.

Delete original:

```
rm file.txt
```

Hard link still works.

4) Soft Link (Symbolic Link)

```
ln -s file.txt softlink.txt
```

Check:

```
ls -l
```

Soft link has different inode.

Delete original → soft link breaks.

3) Process & Job Control (Core OS)

5) Run Process in Background

```
sleep 100 &
```

See jobs:

```
jobs
```

Bring to foreground:

```
fg %1
```

Kill background job:

```
kill %1
```

6) **nice** — Change Priority

```
nice -n 10 sleep 100
```

Check priority:

`ps -l`

Lower nice value = higher priority.

7) **renice**

`sudo renice -5 PID`

Changes running process priority.

4 Memory & CPU Monitoring

8) **free**

`free -h`

Shows RAM usage.

9) **htop** (if installed)

`htop`

Better version of `top`.

5 File Descriptor Concept

10) File Descriptors

Check open files:

`ls -l /etc/passwd`

Check for specific process:

`lsof -p PID`

Standard descriptors:

- 0 → stdin
- 1 → stdout
- 2 → stderr

Test redirection:

```
echo "Hello" 1> out.txt 2> err.txt
```

6 Environment Variables

11) View All Variables

`printenv`

Check specific:

```
echo $PATH
```

Set temporary variable:

```
export MYVAR="OS_LAB"
```

7 Scheduling (Cron Jobs)

Edit cron:

```
crontab -e
```

Example:

Run every minute:

```
* * * * * echo "Hello" >> test.txt
```

Check cron jobs:

```
crontab -l
```

8 Mounting & File Systems

Check mounted drives:

```
mount
```

Disk partitions:

```
lsblk
```

File system type:

```
df -T
```

9 User & Group Management

Create user:

```
sudo useradd testuser
```

Set password:

```
sudo passwd testuser
```

Switch user:

```
su testuser
```

Check groups:

```
groups
```